DataBase Lab 06 (In-Lab Task)

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Task 1

Welcome Page | hr

Worksheet | Query Builder

```
SELECT job_id, AVG(salary) AS avg_salary
FROM employees
GROUP BY job_id
HAVING AVG(salary) > 10000;
```

Query Result

SQL | All Rows Fetched: 6 in 0.043 seconds

	JOB_ID	AVG_SALARY
1	AC_MGR	12008
2	PU_MAN	11000
3	AD_VP	17000
4	FI_MGR	12008
5	SA_MAN	12200
6	AD_PRES	24000

Task 2

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Worksheet | Query Builder

```
SELECT department_id, COUNT(*) AS emp_count
FROM employees
WHERE salary > 2000
GROUP BY department_id
HAVING COUNT(*) > 2;
```

Query Result

SQL | All Rows Fetched: 6 in 0.006 seconds

	DEPARTMENT_ID	EMP_COUNT
1	100	5
2	30	6
3	90	4
4	50	45
5	80	34
6	60	5

Task 3

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Worksheet | Query Builder

```
SELECT department_name
FROM (
    SELECT d.department_name, j.start_date
    FROM departments d, job_history j
    WHERE d.department_id = j.department_id
    ORDER BY j.start_date DESC
)
WHERE ROWNUM = 1;
```

Query Result

SQL | All Rows Fetched: 1 in 0.017 seconds

	DEPARTMENT_NAME
1	IT

Task 4

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Worksheet | Query Builder

```
CREATE INDEX emp_index
ON employees(employee_id);

CREATE INDEX job_index
ON job_history(employee_id);

SELECT employee_id, first_name, last_name
FROM employees
WHERE employee_id NOT IN
      (SELECT employee_id FROM job_history);
```

Script Output | Query Result

SQL | Fetched 50 rows in 0.011 seconds

	EMPLOYEE_ID	FIRST_NAME	LAST_NAME
1	100	Steven	King
2	106	Valli	Pataballa
3	107	Diana	Lorentz
4	108	Nancy	Greenberg
5	109	Daniel	Faviet
6	110	John	Chen
7	111	Ismael	Sciarra
8	113	Luis	Popp
9	115	Alexander	Khoo
10	116	Shelli	Baida
11	117	Sigal	Tobias
12	118	Guy	Himuro
13	119	Karen	Colmenares
14	120	Matthew	Weiss
15	121	Adam	Fripp
16	123	Shanta	Vollman
17	124	Kevin	Mourgos
18	125	Julia	Nayer
19	126	Irene	Mikkilineni
20	127	James	Landry
21	128	Steven	Markle
22	129	Laura	Bissot
23	130	Mozhe	Atkinson
24	131	James	Marlow
25	132	TJ	Olson
26	133	Jason	Mallin
27	134	Michael	Rogers
28	135	Ki	Gee
29	136	Hazel	Philtanker
30	137	Renske	Ladwig

Task 5

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Worksheet | Query Builder

```
SELECT department_id, MAX(salary) AS max_salary
FROM employees
GROUP BY department_id
HAVING MAX(salary) > 2000
ORDER BY max_salary ASC;
```

Script Output | Query Result

SQL | All Rows Fetched: 11 in 0.008 seconds

	DEPARTMENT_ID	MAX_SALARY
1	10	4400
2	40	6500
3	(null)	7000
4	50	8200
5	60	9000
6	70	10000
7	30	11000
8	100	12008
9	110	12008
10	80	14000
11	90	24000

Task 6

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Worksheet | Query Builder

```
SELECT *
FROM (
    SELECT department_id, AVG(salary) AS avg_salary
    FROM employees
    GROUP BY department_id
    HAVING AVG(salary) > 6000
    ORDER BY avg_salary DESC
)
WHERE ROWNUM <= 3;
```

Script Output | Query Result

SQL | All Rows Fetched: 3 in 0.004 seconds

	DEPARTMENT_ID	AVG_SALARY
1	90	16450
2	110	10154
3	70	10000

Task 7

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Worksheet | Query Builder

```
SELECT last_name, job_id
FROM employees
WHERE job_id =
      (SELECT job_id
       FROM employees
       WHERE employee_id = 150);
```

Script Output | Query Result

SQL | All Rows Fetched: 30 in 0.006 seconds

	LAST_NAME	JOB_ID
1	Tucker	SA_REP
2	Bernstein	SA_REP
3	Hall	SA_REP
4	Olsen	SA_REP
5	Cambrault	SA_REP
6	Tuvault	SA_REP
7	King	SA_REP
8	Sully	SA_REP
9	McEwen	SA_REP
10	Smith	SA_REP
11	Doran	SA_REP
12	Sewall	SA_REP
13	Vishney	SA_REP
14	Greene	SA_REP
15	Marvins	SA_REP
16	Lee	SA_REP
17	Ande	SA_REP
18	Banda	SA_REP
19	Ozer	SA_REP
20	Bloom	SA_REP
21	Fox	SA_REP
22	Smith	SA_REP
23	Bates	SA_REP
24	Kumar	SA_REP
25	Abel	SA_REP
26	Hutton	SA_REP
27	Taylor	SA_REP
28	Livingston	SA_REP
29	Grant	SA_REP
30	Johnson	SA_REP

Task 8

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Worksheet | Query Builder

```
CREATE TABLE Job_History1 AS
SELECT *
FROM job_history
WHERE 1 = 2;

INSERT INTO Job_History1
SELECT *
FROM job_history
WHERE end_date = TO_DATE('19-DEC-07','DD-MON-YY');
```

Script Output | Query Result

Task completed in 0.026 seconds

```
Table JOB_HISTORY1 created.


0 rows inserted.
```

Task 9

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Worksheet | Query Builder

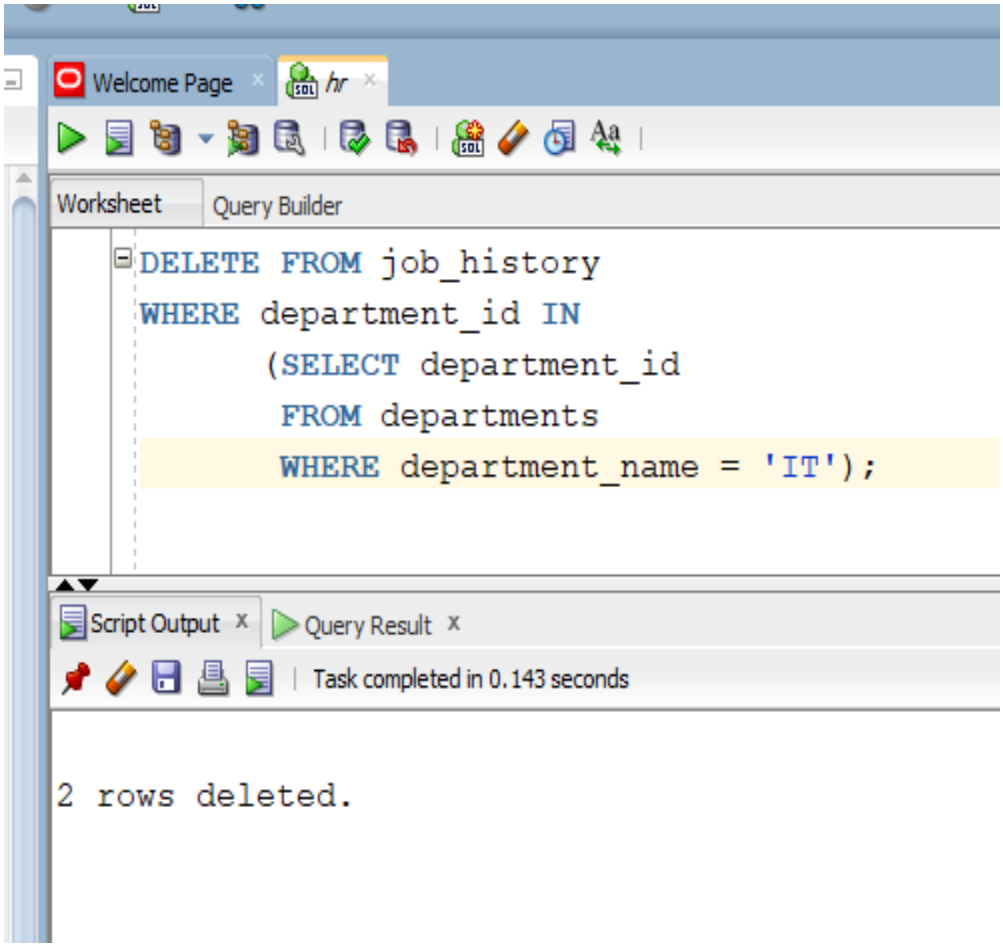
```
INSERT INTO Job_History1
VALUES (101, DATE '2020-01-01', DATE '2020-12-31', 'AC_ACCOUNT', 90);
INSERT INTO Job_History1
VALUES (102, DATE '2019-01-01', DATE '2019-12-31', 'AC_ACCOUNT', 90);
INSERT INTO Job_History1
VALUES (103, DATE '2018-01-01', DATE '2018-12-31', 'IT_PROG', 60);
INSERT INTO Job_History1
VALUES (104, DATE '2017-01-01', DATE '2017-12-31', 'SA_REP', 80);
INSERT INTO Job_History1
VALUES (105, DATE '2016-01-01', DATE '2016-12-31', 'HR_REP', 40);

DELETE FROM Job_History1
WHERE job_id = 'AC_ACCOUNT';
```

Script Output | Query Result

Task completed in 0.028 seconds

```
1 row inserted.

1 row inserted.

1 row inserted.

1 row inserted.

1 row inserted.

2 rows deleted.
```

Task 10


Welcome Page
hr
Worksheet
Query Builder
DELETE FROM job_history
WHERE department_id IN
(SELECT department_id
FROM departments
WHERE department_name = 'IT');
Script Output
Query Result
Task completed in 0.143 seconds
2 rows deleted.